

PROJECT FINAL REPORT
Personalized Networking Assistant

Team ID: _____

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Date: July 2026

1. INTRODUCTION

1.1 Project Overview

Professional networking has become an essential part of career growth, business development, research collaboration, and professional communication. Students, professionals, researchers, and entrepreneurs regularly attend conferences, workshops, seminars, career fairs, and networking events to build valuable professional relationships. However, many participants experience difficulty initiating conversations due to limited preparation, lack of confidence, or insufficient knowledge about the event and discussion topics.

The **Personalized Networking Assistant (Smart Connect AI)** is an Artificial Intelligence-powered web application developed to simplify networking preparation by generating intelligent conversation starters and providing verified information about networking topics. The application analyzes user-provided event descriptions, extracts relevant networking themes, and generates personalized conversation starters that help users engage confidently in professional discussions.

The project integrates **FastAPI** as the backend framework and **Streamlit** as the frontend framework to build a responsive and user-friendly web application. The backend processes user requests, performs event analysis, generates networking questions, retrieves verified knowledge from the Wikipedia API, stores networking session history, and records user feedback. The frontend provides an interactive interface where users can access all functionalities easily.

The application includes modules such as Event Description, Conversation Starter Generator, Knowledge Assistant, Session History, and Feedback Collection. Networking sessions and feedback are stored locally using JSON files, making the application lightweight, efficient, and easy to maintain.

Overall, the Personalized Networking Assistant demonstrates the practical implementation of Artificial Intelligence, Natural Language Processing (NLP), REST APIs, and modern Python web frameworks to improve communication confidence and networking effectiveness.

1.2 Purpose

The purpose of the Personalized Networking Assistant is to provide an intelligent web application that helps users prepare for professional networking events by generating AI-powered conversation starters and delivering verified information using the Wikipedia API.

The project aims to:

- Generate intelligent conversation starters based on event descriptions.
- Improve users' communication confidence.
- Provide verified networking knowledge using Wikipedia.
- Save networking sessions for future reference.
- Collect user feedback.
- Demonstrate FastAPI and Streamlit integration.

- Showcase Artificial Intelligence and Natural Language Processing concepts.
- Develop a scalable networking assistant for future enhancements.

The project serves as an intelligent assistant that enables users to build meaningful professional relationships while improving networking skills and communication effectiveness.

2. IDEATION PHASE

2.1 Problem Statement

Many students and professionals struggle to start meaningful conversations during networking events because they lack confidence, preparation, and relevant discussion topics. Existing solutions provide general networking advice but do not generate personalized conversation starters based on specific events.

The Personalized Networking Assistant solves this problem by analyzing event descriptions, generating AI-powered conversation starters, verifying information through the Wikipedia API, and helping users prepare effectively for networking opportunities.

2.2 Empathy Map Canvas

Says

- I don't know how to start conversations.
- I feel nervous at networking events.
- I need relevant discussion topics.

Thinks

- I want to make professional connections.
- AI can help me prepare better.
- Verified information increases my confidence.

Does

- Searches online before attending events.
- Reads about companies and technologies.
- Uses networking platforms.

Feels

- Nervous before networking.
- Excited after successful conversations.
- More confident with preparation.

Pain

- Lack of conversation ideas.
- Limited event knowledge.
- Communication anxiety.

Gain

- Better networking confidence.
 - Personalized conversation starters.
 - Verified information.
 - Improved professional communication.
-

2.3 Brainstorming

The development team discussed multiple ideas before selecting the Personalized Networking Assistant.

Considered Ideas

- AI Resume Builder
- Career Recommendation System
- Interview Preparation Assistant
- Personalized Networking Assistant

The team selected the Personalized Networking Assistant because it addresses a real-world communication problem and effectively demonstrates Artificial Intelligence, Natural Language Processing, REST APIs, and web application development.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	User Action	Touchpoint	Emotion
Awareness	Wants networking assistance	Web Application	Curious
Preparation	Enters event description	Streamlit Interface	Interested
Processing	AI analyzes event	FastAPI Backend	Hopeful
Conversation Generation	Receives conversation starters	AI Module	Confident
Knowledge Verification	Searches topic	Wikipedia API	Satisfied
Networking	Uses generated questions	Professional Event	Successful

3.2 Solution Requirements

Functional Requirements

- User enters networking event description.
- Generate conversation starters.
- Verify networking topics.
- Save networking sessions.
- Submit feedback.
- Display AI-generated results.

Non-Functional Requirements

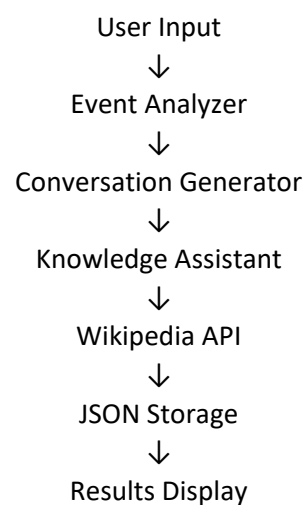
- User-friendly interface.
 - Fast response time.
 - Reliable API communication.
 - Secure local storage.
 - Scalable architecture.
 - Responsive design.
-

3.3 Data Flow Diagram

Level 0

User → Streamlit → FastAPI → Response

Level 1



3.4 Technology Stack

Component	Technology
Programming Language	Python
Frontend	Streamlit
Backend	FastAPI
AI	NLP
API	Wikipedia API
HTTP	Requests
Storage	JSON
Server	Uvicorn
IDE	Visual Studio Code
Version Control	Git & GitHub

4. PROJECT DESIGN

4.1 Problem Solution Fit

The proposed solution addresses the challenge of networking preparation by combining Artificial Intelligence, NLP, REST APIs, and verified knowledge retrieval into a single web application.

4.2 Proposed Solution

The application analyzes networking events, generates conversation starters, verifies networking topics using Wikipedia, stores networking history, and collects user feedback through an interactive Streamlit interface powered by a FastAPI backend.

4.3 Solution Architecture

The architecture consists of:

- User Layer
- Streamlit Frontend
- FastAPI Backend
- Event Analyzer

- Conversation Generator
 - Knowledge Assistant
 - Wikipedia API
 - JSON Storage
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5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project was divided into multiple sprints covering:

- Environment Setup
 - Backend Development
 - Frontend Development
 - API Integration
 - Knowledge Assistant
 - Session History
 - Feedback Module
 - Testing
 - Documentation
-

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The application was tested for:

- API Response
- Conversation Generation
- Knowledge Retrieval
- Session Saving
- Feedback Submission
- UI Responsiveness
- JSON Storage
- FastAPI Communication

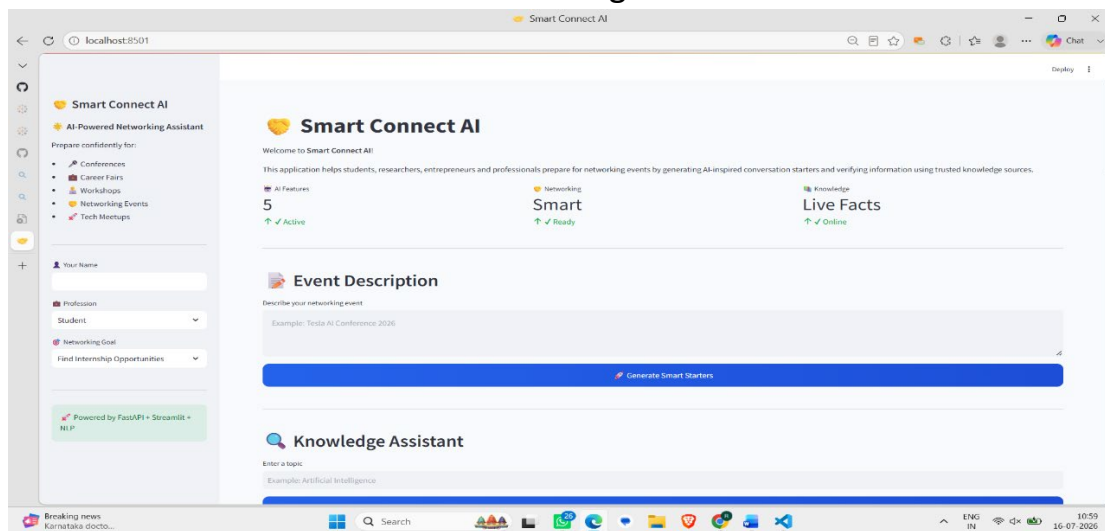
All tests were successfully completed.

7. RESULTS

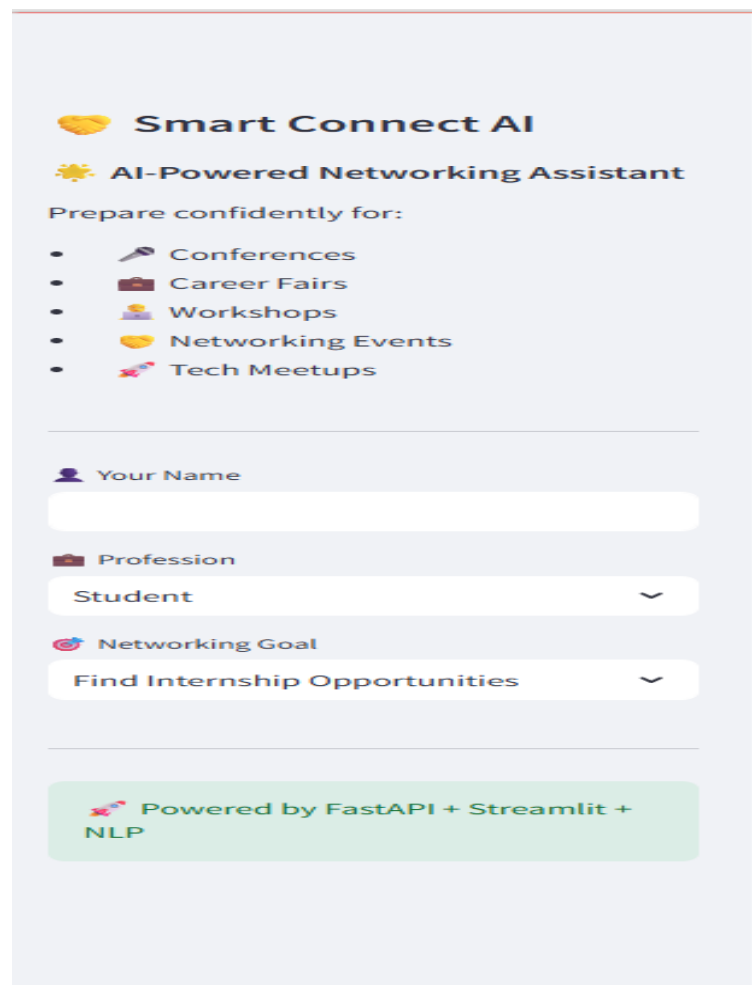
7.1 Output Screenshots

Include screenshots of:

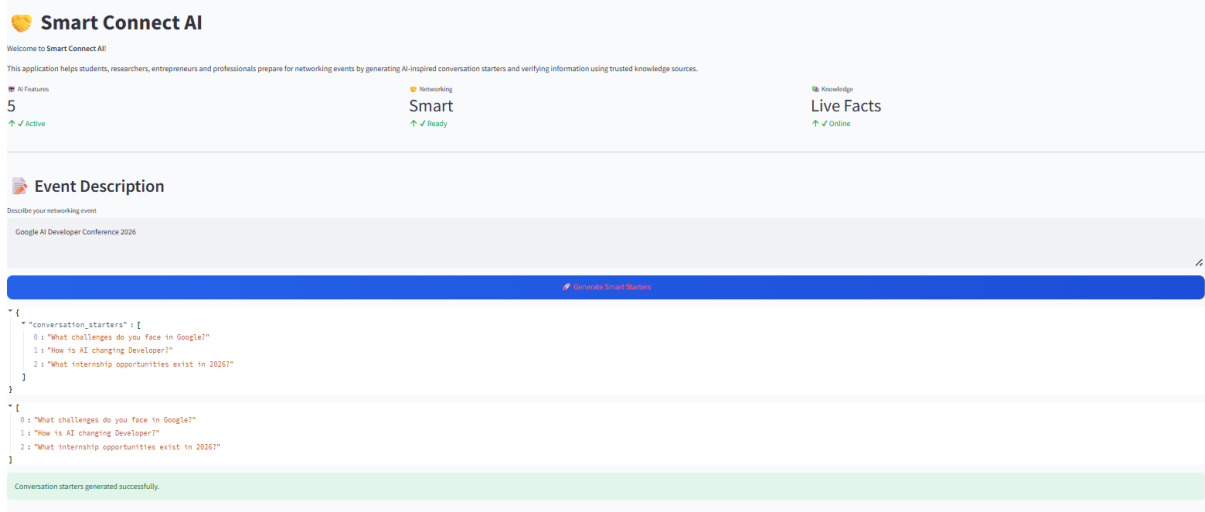
Home Page



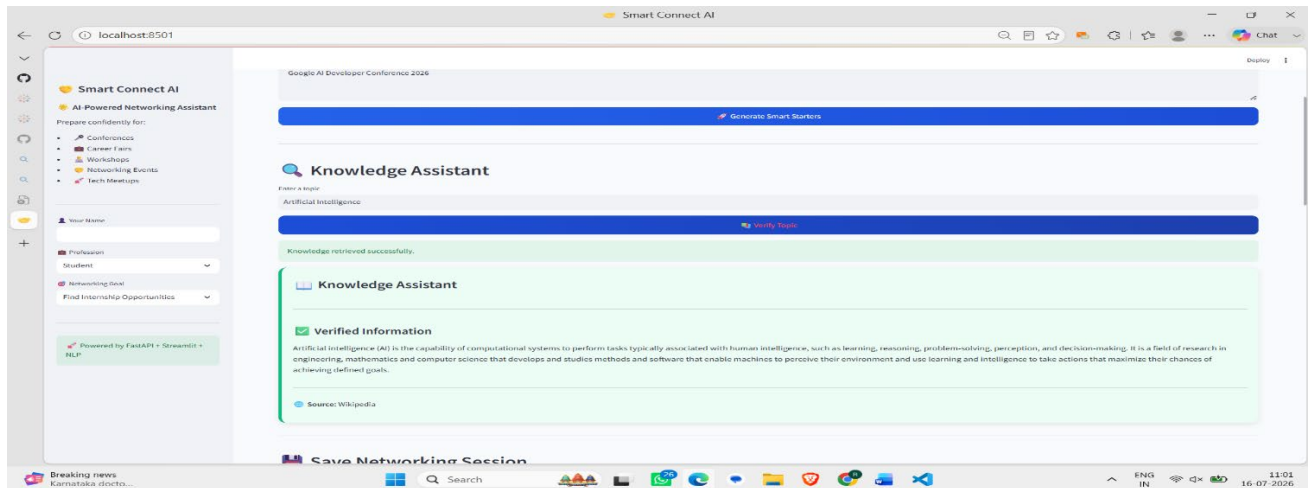
Sidebar



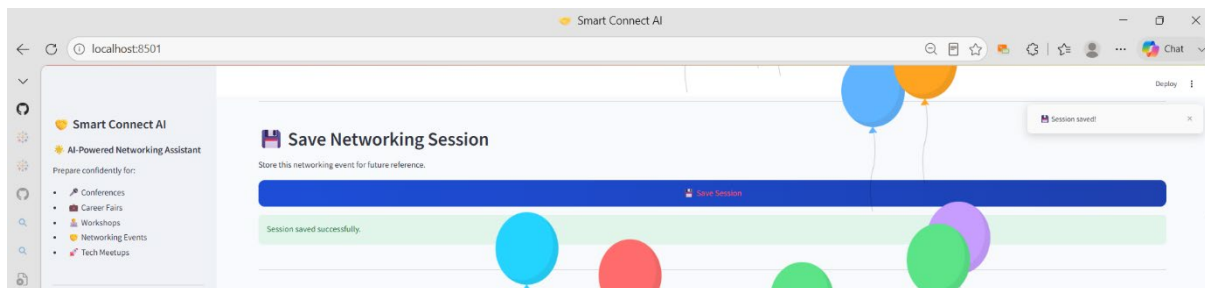
Event Description



Knowledge Assistant



Save Session



Feedback Form

The screenshot shows the 'Feedback Form' section of the 'Smart Connect AI' application. The interface includes a sidebar with navigation options like 'Conferences', 'Career Fairs', 'Workshops', 'Networking Events', and 'Tech Meetups'. The main content area has a 'Save Networking Session' button and a 'Feedback' section with a text input field. A 'Submit Feedback' button is visible, and a green message bar at the bottom indicates 'Feedback submitted successfully.' The application is running on localhost:8501.

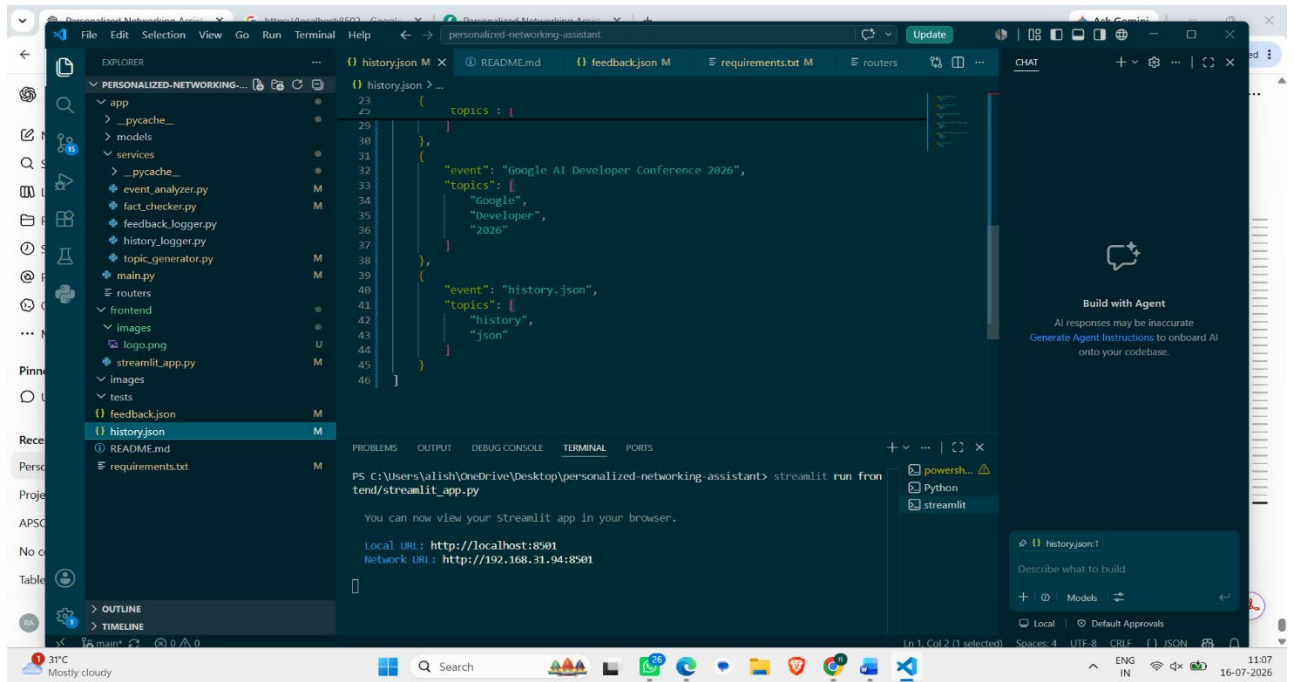
Project Summary

The screenshot displays the 'Project Summary' page. It features a 'Key Features' section with a list of capabilities: Event Topic Analysis, AI Conversation Starters, Knowledge Assistant, Session History, Feedback Collection, FastAPI Backend, and Streamlit Frontend. A 'Technology Stack' section lists Python, FastAPI, Streamlit, Wikipedia REST API, JSON Storage, and REST API. The 'Project Highlights' section shows 7 features, with 1 complete and 6 working. The application is powered by FastAPI and Streamlit, and is running on localhost:8501.

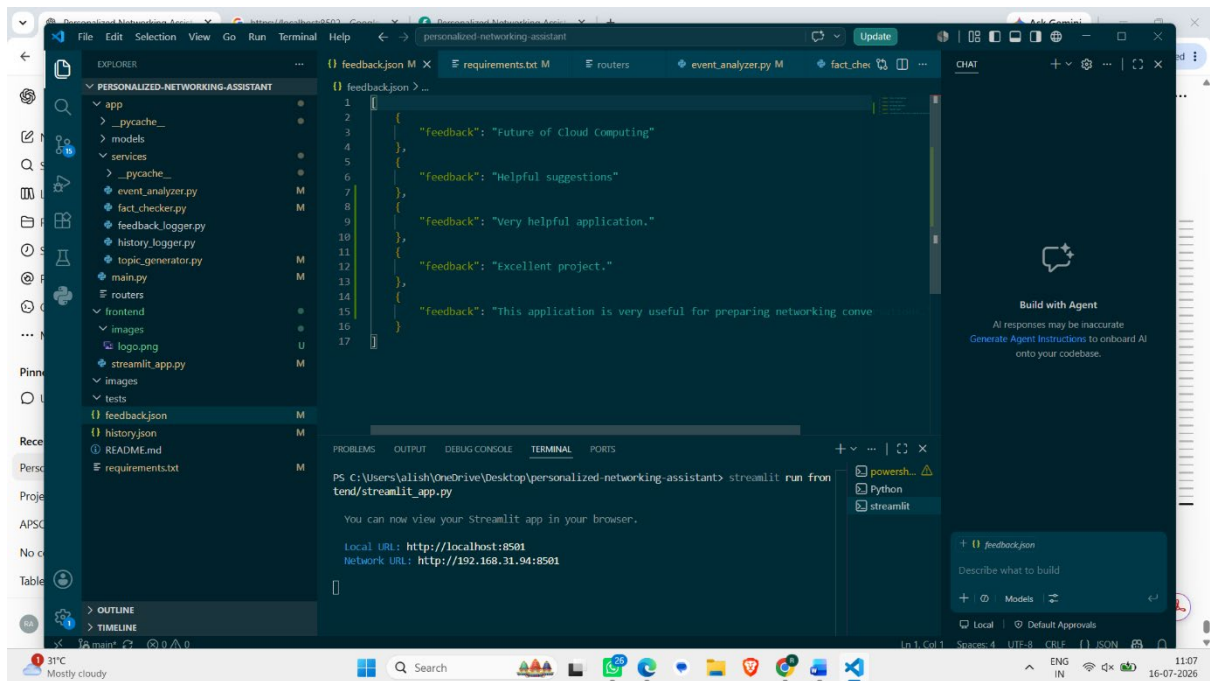
FastAPI Swagger

The screenshot shows the FastAPI Swagger UI for the 'Personalized Networking Assistant'. The interface displays a list of endpoints under the 'default' tag, including GET /home, GET /analyze/analyze, GET /generate/Generate Conversation, GET /fastcheck/Fastcheck Feedback, GET /save-history/history, and GET /feedback/Feedback. The 'Schemas' section shows the HTTPValidationError and ValidationError objects. The application is running on 127.0.0.1:8000/docs. The bottom of the image shows a Windows taskbar with the date 16-07-2026 and time 11:04.

history.json



feedback.json



8. ADVANTAGES & DISADVANTAGES

Advantages

- AI-powered networking preparation.
- Personalized conversation starters.
- Verified information using Wikipedia.
- User-friendly interface.
- Fast API communication.
- Lightweight architecture.
- Local data storage.
- Easy to extend.
- Modular design.
- Scalable application.

Disadvantages

- Requires internet for Wikipedia API.
 - Local JSON storage only.
 - No authentication.
 - Limited AI model.
 - No cloud deployment.
-

9. CONCLUSION

The **Personalized Networking Assistant (Smart Connect AI)** successfully integrates Artificial Intelligence, FastAPI, Streamlit, Natural Language Processing, and the Wikipedia API into a single web application that assists users in preparing for networking events. The application generates intelligent conversation starters, retrieves verified information, stores networking sessions, and collects user feedback through an interactive interface. The project demonstrates modern web development practices and provides a scalable foundation for future AI-powered networking solutions.

10. FUTURE SCOPE

- GPT integration
- LinkedIn integration
- Voice Assistant
- Firebase Authentication

- Cloud Database
 - Mobile Application
 - Multi-language Support
 - Recommendation System
 - Speech Recognition
 - Real-time Event Analysis
-

11. APPENDIX

Source Code

Python source code developed using:

- FastAPI
- Streamlit
- Requests
- Wikipedia API
- JSON

Dataset

No external dataset was used. The application processes user-provided event descriptions and retrieves verified information dynamically through the Wikipedia API.

GitHub Repository: -

<https://github.com/Alisha40477/Personalized-Networking-Assistant>

Project Demo: -

<https://drive.google.com/drive/folders/1HDyutqDbBQzW2F01GboAx08L70N4zn1j>
